

Printers Test

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In the early days of personal computing, printers and scanners were the chosen peripherals one would buy along with their PC. Then, printers were expensive and the consequent running cost would limit their use to occasional printing and very few could afford them. But today the prices have fallen and one can easily buy a small printer for not more than Rs 3,000. Also, Inkjet printers were the only option because Laser technology was new and laser printers would cost five times more than a similar spec inkjet printer. Today a good laser printer can be had for not more than Rs 5,000.

Printing technology has made rapid advancement in the last couple of years, and today they boast of mind-numbing speeds and photolab printing quality, irrespective of whether you choose Inkjet or Laser. Inkjet printers were typically known to be better at producing

Laser and inkjet printers tested across two categories to help you find the right solution for your business

coloured printouts at lower speeds. Laser printers were known to deliver mono prints at impressive speed and proved economic in the longer run. But now you get Inkjet printers which offer similar benefits as laser printers and vice-versa.

Volume of printing should be used to decide the type of printer except in some special cases. Laser printers work out cheaper when print volumes are high—100 pages a day. On the other hand, for 10 pages a week, an entry level inkjet printer is sufficient.

In this test we are looking at both laser as well as inkjet printers, but not exceeding Rs 20,000. The Inkjet printers have been divided into two distinct classes. Products for home and small offices and photo printers with loads of features. For lasers, there are two categories as well: for personal lasers, we have grouped all laser printers below 10,000. Then there's the small workgroup laser printers priced between 10K to 20K, for

SoHo applications. We will begin with the laser category first and then look into the inkjet products.

Laser Printers

Personal Laser Printers

As the name suggests, these are laser printers for personal use. In this category we got nine printers priced between Rs 4,000 and 10,000. For clarity, we have further divided them into two categories.

Brother's HL-2140, Canon's LBP 2900B, Samsung's ML-1640 and Xerox's 3117 printers are priced below Rs 6,000, and form the entry-level personal laser group.

Features

A personal laser will often be kept near your desk and hence it has to look good. Canon's LBP 2900B excels here due to its intelligent design and compact form factor. Samsung's ML-1640 has a familiar design, but the proportions are well within acceptable limits.

Xerox 3117 replicates Samsung's design, but is finished in white. The HL-2140 is a new series from Brother and is designed to appeal with its low profile, horizontal layout—but it uses more desk space.

Laser printers are often defined by their speed given as PPM (pages per minute). All these devices nearly delivered on their speed promises. Importantly, each device was able to print the first page within 10 seconds and that matters more than PPM. Except for Canon, all other printers are equipped with 8 MB of onboard memory. But despite lower memory, the Canon machine is just as fast when printing the first page. Canon uses system memory, a method it calls CAPT, and it's definitely delivering the goods.

Canon's LBP 2900B, Samsung's ML-1640 and Xerox's 3117 had paper trays that tuck away into the printer when the flaps are closed. However, Brother's HL-2140 has a paper cassette which holds paper, due to this design the Brother printers seem to occupy more space. You have to remember that in the first three printers, the paper trays extend from the body, and this uses nearly the same space as the Brother HL-2140 does. Hence, all four printers irrespective of their design need same amount of desk space.

Eco mode is available on all these printers and can be set via the driver settings. In the Eco mode, the printer uses less toner and hence less electricity too. Automatic dual-side printing is not offered on any of these printers. The drivers provided with each of these printers offered extensive control over the device.

The usual options for paper, orientation, quality were present and other pre-defined options for printing text, graphics, photos were present for one touch selection.

Performance

On printing speed, Canon's LBP-2900B was ahead of other printers on most of the tests we put it through. The first printout is delivered in just 10 seconds, while the rest take 11 seconds. When printing large files such as the combi-document, the

CAPT technology helps the Canon maintain the lead. Both Samsung and Xerox deliver almost identical performance figures, with the Brother HL-2140 trailing slightly behind.

If we scrutinise print quality, then Brother's HL-2140 delivered excellent results, followed by Samsung and Xerox. Canon's LBP-2900B missed out on some details especially in the lighter shades of the grey scale. Graphs are faithfully reproduced by the Brother, Samsung and Xerox printers, and Canon does a good job, but misses out on finer details. With respect to laser printers tested a couple of years ago, all the four printers here are remarkably better in terms of print quality—especially in the graphics department. The result is smooth consistent-looking images, without any gradient or pixilation.

If you are on a shoestring budget, and want a laser printer, then go for Xerox 3117. Both the Samsung

ML-1640 and Brother HL-2140 are excellent buys—just make sure your dealer readily stocks their toners. We have chosen Brother's HL-2140 as the *Best Buy* over Samsung's ML-1640 simply for its better performance and bigger paper tray. If Brother Printers are not available in your locality, then the Samsung ML-1640 should be considered.

Brother's HL-2150N, Canon's LBP 1210, HP's P1007 and P1008 and Samsung's ML-2245 are all priced between Rs 6,000 and 10,000, and they form our next category of premium personal laser printers. These machines offer higher duty cycles, and hence can take some thrashing. They also offer some additional features such as networking options and higher speeds.

Features

The P1007 and P1008 are new laser printers from HP and sport a modern design. The Brother HL-2150N is identical to the HL-2140

How We Tested

Our test-bed comprised of a Pentium 4 3.2 GHz processor, plugged to an Intel 875PBZ motherboard, with 512 MB of corsair 400 MHz DDRRAM, MSI Ti4800 display card and 40 GB 7200 rpm ATA IV Seagate Barracuda hard drive. We used Windows XP with SP1 as our chosen operating system. The system was made up to date by loading all the latest chipset drivers and graphics driver. The USB patch for windows XP was also installed and the devices were connected on the USB 2.0 port for optimum performance. We used 100 gsm papers for doing all test except for the drain test where we used 75 gsm papers. For printing photographs, we used NovaJet's 130 g/m² inkjet photo paper across all the printers, since they are widely available across the country.

We tested all the printers on the same test benchmarks which comprised of a text document, a combi document, a continuous document and an image file. These test documents incorporate every small detail that a printer will need to reproduce right from smallest point text, lines, specular effect (shining) on metal, etc.

The combi document has many sections with text of varying points; lines draw in a circle, seven colour bars of decreasing gradient and a small picture at one corner.

Each of these subsection tests the printer's ability to reproduce every detail exactly as seen in the original file. The varying text point size test the printers ability to reproduce even the smallest possible text, in other word it test the spatial resolution of the printer when it comes to printing text. Where as the circle of lines test the printer's ability to reproduce lines without jarring the edges. Similarly the seven colour bar test the colour reproduction capabilities of the printer cartridges.

The continuous document is a small report with bar graphs, pie charts, line graphs, images and test all spread over six pages. We choose this document as it resembles the real world usage in most cases.

The image document is made up of many elements, with varying contrast and brightness areas. This document test the printers colour reproduction ability on a photo paper, so that the final output looks as striking and brilliant as that taken from a studio.

All the test files were printed for two different quality settings, except for the test image. The text and the combi-document file were printed at Normal and Best quality settings, where as for the image file we used only the best quality settings.